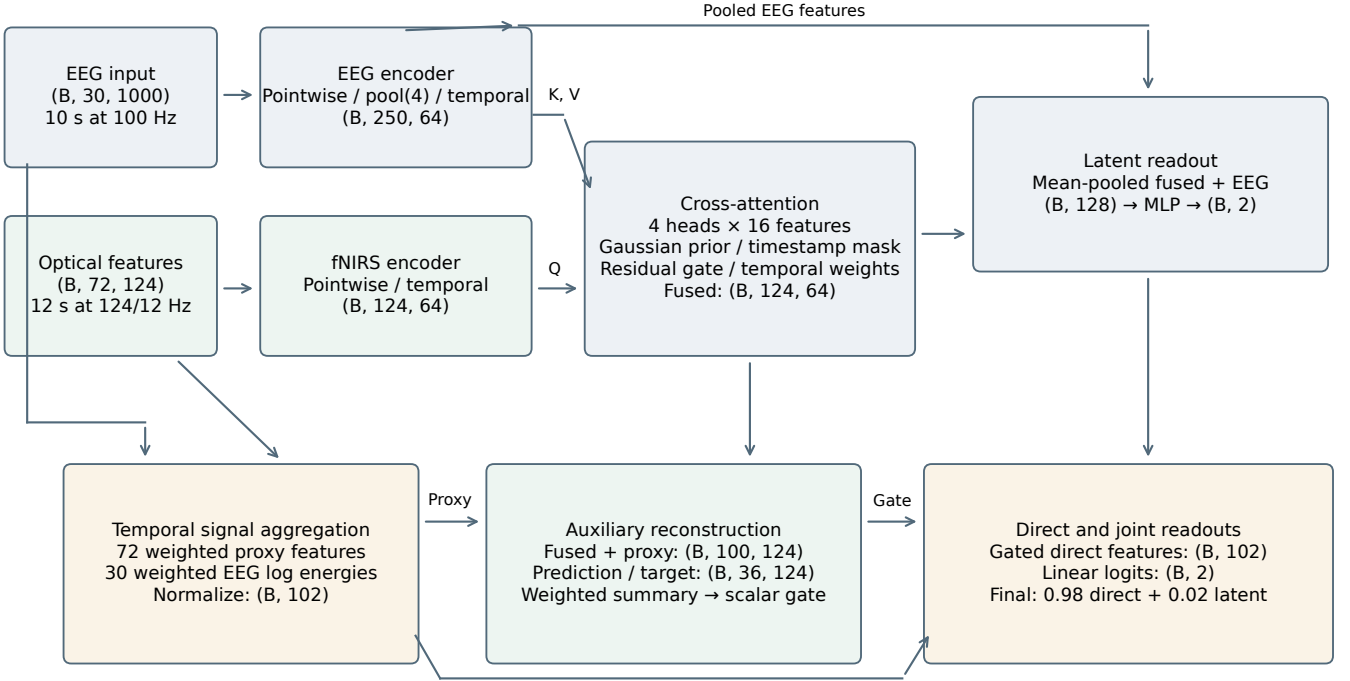

Pooled EEG features
EEG input
(B, 30, 1000)
10 s at 100 Hz
EEG encoder
Pointwise / pool(4) / temporal
(B, 250, 64)
K, V
Optical features
(B, 72, 124)
12 s at 124/12 Hz
fNIRS encoder
Pointwise / temporal
(B, 124, 64)
Q
Cross-attention
4 heads × 16 features
Gaussian prior / timestamp mask
Residual gate / temporal weights
Fused: (B, 124, 64)
Latent readout
Mean-pooled fused + EEG
(B, 128) → MLP → (B, 2)
Temporal signal aggregation
72 weighted proxy features
30 weighted EEG log energies
Normalize: (B, 102)
Proxy
Auxiliary reconstruction
Fused + proxy: (B, 100, 124)
Prediction / target: (B, 36, 124)
Weighted summary → scalar gate
Gate
Direct and joint readouts
Gated direct features: (B, 102)
Linear logits: (B, 2)
Final: 0.98 direct + 0.02 latent